2. Literature Review

Project: Software-Defined Industrial Visual Anomaly Detection System: Cloud MLOps Pipeline & Simulation
Supervisor: Randa Hesiam

Section 2.1 contains the review feedback and overall evaluation; fields still marked TBD (reviewer, review date, team response) are to be completed after the supervisor review. Section 2.2 lists improvements the team already identified in the proposal. Section 2.3 is a template to be filled from the official criteria.

2.1 Feedback & Evaluation

Item	Details
Reviewer (lecturer / supervisor)	TBD
Review date	TBD
Documents reviewed	Project Proposal, Project Plan, Risk Assessment, KPIs

Reviewer Comments

#	Area	Comment	Severity (Minor / Major)	Status
1	Early Integration Testing	Emphasize the creation of mock/placeholder ONNX models during Week 2 to allow you to integrate the AWS Lambda and API layers without waiting for final model convergence.	Minor	Open
2	Proactive Cold-Start Management	Given the constraints of SageMaker Serverless endpoints, strict adherence to the planned EventBridge 5-minute keep-warm schedule and pre-demo parallel warmup scripts is mandatory to guarantee a flawless live defense presentation.	Major	Open
3	Scope Guarding on MLOps Orchestration	Maintain the current decision to use a scripted, end-to-end MLOps retraining workflow (Python script triggered via EventBridge) rather than over-engineering a full SageMaker Pipelines orchestration, ensuring on-time delivery.	Minor	Open

Team Response

#	Action taken	Owner	Date
1	TBD	TBD	TBD
2	TBD	TBD	TBD
3	TBD	TBD	TBD

Overall Evaluation

The project proposal demonstrates an exceptional, industry-grade architecture for an automated industrial visual anomaly detection system. By shifting away from physical hardware dependencies toward a robust software-defined simulation paired with an event-driven AWS serverless stack (S3, Lambda, SageMaker Serverless, DynamoDB), the team has successfully scoped a highly feasible and impactful Capstone project. The choice of YOLOv8 optimized via ONNX Runtime guarantees meeting the sub-second latency target (<1s), while PatchCore serves as a solid offline unsupervised benchmark. The inclusion of rigorous preprocessing modules (preprocess.py), automated mask-to-box conversion logic, and an explicit MLOps drift-retraining loop reflects a mature engineering mindset. The 6-week timeline is tight but well-mitigated through parallel milestone structures and clear pair-programming role assignments.

2.2 Suggested Improvements

Improvements and extensions identified by the team during planning. They are outside the six-week scope but are candidates for future work or for the supervisor's feedback.

#	Area	Suggested improvement	Reason
1	Materials	Add a glass material profile	Glass is out of scope in this version; the material-profile design allows adding it without changing the architecture
2	Data	Validate on factory-captured images from a real production line	Data is public and partly lab-captured, so results show the method, not guaranteed line performance
3	Detection	Add rotated-box (OBB) detection for long diagonal defects	Axis-aligned boxes contain mostly background for thin diagonal cracks
4	Unseen defects	Combine YOLOv8 with an unsupervised detector (e.g. lighter PatchCore variants) as a second opinion	YOLOv8 cannot detect defect types absent from its training data
5	MLOps	Port the scripted retraining workflow to SageMaker Pipelines	Gate and registry metadata are already compatible
6	MLOps	Automate more of the human-in-the-loop labelling/approval steps	Retraining is currently manual at review, labelling and approval
7	Latency	Provisioned Concurrency or a lighter model/imgsz trade-off study	Reduces cold-start impact beyond the keep-warm approach
8	Dashboard	User accounts, roles and alert subscriptions	Needed for real multi-user factory use
9	Evaluation	Real user study for adoption and usability	Adoption cannot be measured with a simulated line
10	Supervisor input	TBD	To be added after review

2.3 Final Grading Criteria

No official grading distribution from the faculty has been provided yet. The table is left blank to be filled from the official criteria.

Category	What is assessed	Weight (%)	Marks
Documentation	Proposal, project plan, requirements, final report, runbooks, dataset register	TBD	TBD
Implementation	Preprocessing and box-conversion modules, model, AWS pipeline, simulator, dashboard, MLOps loop	TBD	TBD
Testing	Unit tests, ONNX parity check, held-out test accuracy, latency and load tests, KPI acceptance	TBD	TBD
Presentation / Defense	Presentation, live demo, handling of questions, fallback readiness	TBD	TBD
Other (if any)	TBD	TBD	TBD
Total		100	TBD

Source of criteria: TBD (faculty regulation / supervisor)